

activities. The use of Raspberry Pi clusters is an economical solution for testing. Engineers use these systems to duplicate actual IoT systems, which allows them to monitor device performance under various operational scenarios. They perform tests that duplicate network breakdowns, device shutdowns and sensor failures to study system responses.

Through GitHub, users can access small projects that demonstrate Arduino board hardware fault testing. The experiments enable developers to create IoT systems that maintain operational reliability through fault occurrences.

Currently, there is no open source framework specifically for chaos engineering in IoT devices. Without such ‘chaos agents’ on microcontrollers, it is difficult to run controlled system failures, create sensor errors and then automatically recover the system. A framework that includes pre-built test protocols for different IoT platforms would make experiments much easier. A specialised toolkit for edge devices would allow safe testing without damaging actual hardware.

The development of this toolkit would introduce substantial improvements to the current field of operations. A specific chaos engineering system for IoT devices would transform edge device development and maintenance practices just as Kubernetes transformed cloud infrastructure

management. The framework would improve the operational safety and reliability of smart home equipment and industrial sensors, and enable them to operate effectively under actual operational conditions.

Challenges, risks and ethical questions

While chaos engineering helps improve IoT system reliability, it also comes with security risks. In addition, it raises multiple ethical concerns. Unlike cloud servers, IoT devices can be permanently damaged if something goes wrong. For example, a faulty firmware update or a poorly timed test can make a device completely unusable.

Real-world events demonstrate the high level of danger involved. The 2021 software update for Tesla vehicles resulted in short-term failures of the autopilot system. The system failure generated safety hazards for drivers operating their vehicles. The operation of medical devices including insulin pumps and pacemakers depends on dependable firmware systems. Software failures in medical devices could threaten thousands of patients throughout the world.

Safety is one of the most important concerns in IoT chaos engineering. Testing connected cars, medical implants and industrial robots can be dangerous to humans. Running these tests in real production environments is usually restricted. Even simulated tests

can create potential security risks if not managed carefully.

The IoT ecosystem is also very complex, which adds to the difficulty. IoT devices use a wide range of hardware, operating systems and communication protocols. This variety makes it hard to design chaos tools that work across all systems. The lack of a single unified platform means experiments are more difficult. Each fragmented system increases the chance of unforeseen failures.

Personal data protection remains another important concern. The execution of chaos experiments results in accidental exposure of confidential information. For example, the testing of faulty smart home cameras and health monitors may lead to privacy breaches as these devices store private video footage and individual health information. The identification of legal and ethical obligations for these system failures becomes extremely challenging. The responsibility for these failures lies with engineers who run tests, the companies that operate devices, and the developers who built testing tools.

Organisations need to maintain ethical standards when they use chaos engineering to safeguard their IoT systems. Engineers who want to perform IoT chaos testing need to follow established safety protocols. The IEEE’s ‘Ethically Aligned Design’ for autonomous systems together

Table 3: Open source chaos engineering tools and IoT challenges

Aspect	Description	Outcome
Cloud tools	LitmusChaos, Chaos Mesh, Gremlin	Run controlled failures, improve system reliability
Cloud applications	Kubernetes managed servers and containers	Observe app behaviour during failures
IoT challenges	Small OS, microcontrollers, limited memory	Standard cloud tools not compatible
Testing methods	Raspberry Pi clusters, Arduino fault tests	Simulate failures, monitor device performance
Research findings	Sensor errors, fast battery depletion	Identify weaknesses, improve resilience
Current gap	No dedicated open source IoT chaos framework	Hard to run controlled failures safely
Proposed solution	Specialised chaos toolkit for edge devices	Enable safe testing, improve reliability
Importance	Failure-tolerant operations for billions of devices	Ensure IoT systems operate under real conditions